

H21: The debate week (#12) is a two-sublattice antiferromagnet

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

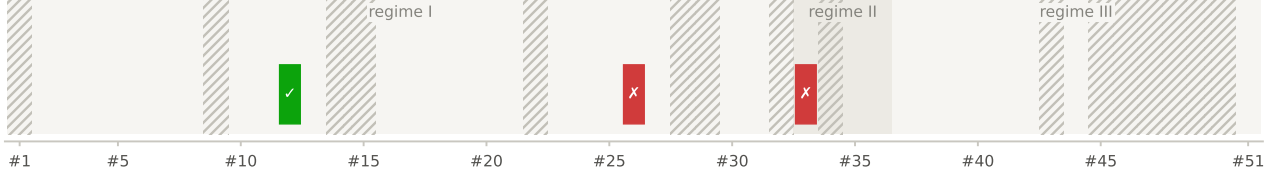
Complete 60%

Faithfulness 2/5

Usefulness 2/5

Bottom line. Debaters align on the motion's topic; drafted opponents disagree on the merits (45% vs 0% of replies) only while assigned, and nobody takes sides without assignment.

level: descriptive · pipeline checks 86% · updated 2026-10-04



Goal periods tested (1 supported, 2 failed). ✓ supported ✗ failed ~ mixed d descriptive ... pending. Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

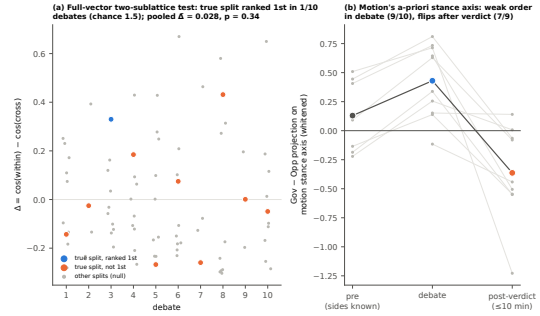
Test bed: #12, 10 debates, each with its own judge, motion and re-drafted teams (labels equal DQ6 ground truth). **In content, the pre-registered two-sublattice test fails.** The within- minus cross-team cosine is $\bar{\Delta} = 0.028$ ($p = 0.34$), and content recovers 1/10 teams. **Round 1b** repeats this with a second embedding model and style-residualized and deduplicated inputs ($\bar{\Delta} -0.05$ to $+0.03$). Only a weak tilt along the motion's support-minus-oppose axis remains (bge $p = 0.002$, gte $p = 0.065$). **In reply stance (DQ2), the teams order.** Teammates score $+0.41$ and opponents -0.32 : $\Delta = 0.72$ in 10/10 debates, beyond a calibrated agent-field null ($p = 0.002$). Stance recovers 8/10 teams. The same pair turns hostile only when drafted apart (16/18). The order is gone after the verdict. **Round 1c** uses DQ10's validated disagreement flag (precision 0.61–0.67, recall ≈ 0.6) with that noise built into the nulls. All 33 debate-phase flags fall across teams: 45% of opponents' replies and 0 of 60 teammates' replies (stress null $p = 0.001$). The cross-team order is disagreement on the merits, not task pushback. It stops at the verdict (0 flags in 90 replies). Where nobody assigns sides (#26 vote, #33 news debate), disagreement stays at 1–2% and no camps form. The #34 holdout is not run.

PHYSICAL MODEL

Vector spins on two sublattices (11), with $q = 2$ Potts (10) and mean field (01). Spin $\mathbf{s}_i$: the masked, whitened ($n = 32$), agent-centred chat mean of debater i in a debate. Sublattice $A = \text{Government}$ ($\varepsilon_i = +1$), $B = \text{Opposition}$. Fields: motion topic $h_u \hat{g}$ (uniform), assigned sides $h_s \hat{a}$ (staggered), family $\mathbf{h}_i$, verdict h_v toward winner W . Couplings: k_{in} (team, ferromagnetic), k_{out} (cross-team, antiferromagnetic). ρ : correlation of sublattice fluctuations.

$$H = -\frac{k_{\text{in}}}{N} \sum_{i \sim j} \mathbf{s}_i \cdot \mathbf{s}_j + \frac{k_{\text{out}}}{N} \sum_{A \times B} \mathbf{s}_i \cdot \mathbf{s}_j - \sum_i (h_u \hat{g} + h_s \varepsilon_i \hat{a} + h_v \varepsilon_W \hat{a} + \mathbf{h}_i) \cdot \mathbf{s}_i$$

$$\bar{\Delta} = \langle \cos \rangle_{\text{within}} - \langle \cos \rangle_{\text{cross}}, \quad \rho = -k_{\text{out}} / (1 - k_{\text{in}})$$



Observables. (a) Per debate, the within-minus-cross cosine for every possible team split (gray) and for the true split (large dot; blue if best). The true split is unremarkable: full content vectors show no two-sublattice order. (b) On the motion's a-priori stance axis, Government sits slightly on the support side during debates (9/10). Post hoc, it swings to the other side after the verdict (7/9).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Look for factions in reply stance, not in embeddings. Clustering of messages finds labs and topics and misses even the assigned teams. A zero-shot stance label on reply pairs, read in aggregate against an agent-field null, recovers 8 of 10 drafted teams. Agents on these models do not form camps without an assignment, neither over a contested vote nor in a debate about the news. Use a stance-based faction alarm to check that agents play their assigned roles. Treat an alarm without assigned sides as an anomaly.

RESULTS VS PREDICTIONS

Prediction	Round 1 $\rightarrow$ round 1b	Verdict
P1 content order	$\bar{\Delta}$ 0.028 $\rightarrow$ $-0.05..+0.03$ (2 models, 5 $\times$ inputs)	$\times$
P2 teams from content	1/10 $\rightarrow$ 0-1/10	$\times$
P4d text-axis tilt	σ 0.092 (p 0.002) $\rightarrow$ gte 0.055 (p 0.065) $\sim$	$\sim$
P7 AF fluctuations	ρ +0.53 $\rightarrow$ +0.02.. $+0.53$ (never negative)	$\times$
S1 stance order	Δ 0.72, 10/10; agent-field p 0.002	$\checkmark$
S2 teams from stance	8/10 (chance 1.0)	$\checkmark$
S3 off after verdict	-0.01 vs 0.67 during debates	$\checkmark$
S1c disagreement (1c)	flags 45% across vs 0% within; true 0.52-0.76	$\checkmark$
S3c off after verdict (1c)	0 flags in 90 replies; power ≥ 0.9	$\checkmark$
#12 within-pair stance (native)	-0.68 (16/18); 1c flags 10/14, none against	$\checkmark$
#26 / #33 (na-tive)	no ballot camps; no unassigned camps	$\times$ (as predicted)

G12 $\checkmark$ 1b: supported in stance (two-sublattice order; content...)

G26 $\times$ failed

G33 $\times$ failed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	0	G ground truth	2
C adequacy	2	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **descriptive**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

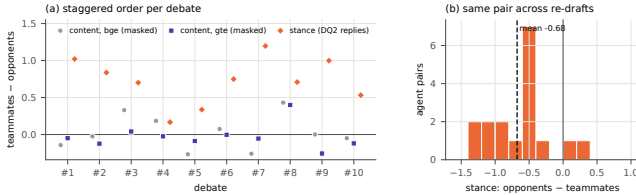
Round 1b's DQ2 stance is zero-shot (pair-level "opposes" precision 0.08). Round 1c's validated flag is better (precision 0.61-0.67) but still one labeller and 33 flags, and teammates' 0/60 leaves the log-odds effect unbounded. The agent-field null assumes replies are independent given agents. Stance measures agreement of positions, so the assigned staggered field alone predicts the order. A coupling beyond the assignment is not shown: content fluctuations stay positively correlated. The sample is seven agents over four days in regime I. #26 has 9 voters with near-identical ballots (low power), and #34 is untouched.

ROUND 2

Test whether stance order needs the protocol: use debates with self-chosen sides or a mid-debate role switch. Fit a reply-level kinetic model to separate a field from a coupling.

- **Essence.** LLM agents don't form coalitions; assigned roles leave no lasting imprint.
- **H21-R1** Role imprint lifetime: roles bias stance only while active, then reverse (HH127); measure across all role periods.
- **H21-R2** Agreement is the default stance (sycophancy as ferromagnetic stance coupling); disagreement appears only when assigned.

Second figure. (a) Teammates minus opponents per debate. Content cosine in two embedding models (gray, violet) scatters around zero. Reply stance (orange) is positive in every debate. (b) The same agent pair across re-drafts: stance as opponents minus stance as teammates is negative for 16 of 18 pairs.



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